

| SEQ NO                                                                             | REV | NOTE NO. | ITEM NAME                                      | ITEM CODE | SIZE RANGE / SMALL END SIZES                                                                                                                                     |      | DESCRIPTION                                                                                    | REFERENCE / LARGE END SIZES |                                                                                                                                    | NOTES |                                                                                                                                         |    |            |            |                   |                |        |
|------------------------------------------------------------------------------------|-----|----------|------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------|----|------------|------------|-------------------|----------------|--------|
| 10                                                                                 |     |          |                                                |           |                                                                                                                                                                  |      | ***** VALVES *****                                                                             |                             |                                                                                                                                    | 810   | NIPPLE TO BE CUT FROM SCH 160 PIPE.                                                                                                     |    |            |            |                   |                |        |
| 20                                                                                 |     | VB2      |                                                |           |                                                                                                                                                                  |      | ***** BALL VALVES *****                                                                        |                             |                                                                                                                                    | 823   | USE OF THIS ITEM IS LIMITED TO MAXIMUM DESIGN TEMPERATURE OF 250°C                                                                      |    |            |            |                   |                |        |
| 30                                                                                 |     |          |                                                |           |                                                                                                                                                                  |      |                                                                                                |                             |                                                                                                                                    | B1    | ALL BOLTS ARE METRIC DIA.                                                                                                               |    |            |            |                   |                |        |
| 30                                                                                 | U04 |          | BALL                                           | A01AC01   | 3                                                                                                                                                                | 4    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO                                 | 0                           | 0                                                                                                                                  | B2    | 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10                                                 |    |            |            |                   |                |        |
| 40                                                                                 | U04 |          | BALL                                           | A01AC02   | 3                                                                                                                                                                | 4    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO                                | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 50                                                                                 | U04 |          | BALL                                           | A01AC03   | 6                                                                                                                                                                | 6    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB GO                                 | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 60                                                                                 | U04 |          | BALL                                           | A01AC04   | 6                                                                                                                                                                | 6    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB GO                                | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 70                                                                                 | U04 |          | BALL                                           | A01AC07   | 8                                                                                                                                                                | 8    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO                                 | 0                           | 0                                                                                                                                  | B6    | 7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22                                                 |    |            |            |                   |                |        |
| 80                                                                                 | U04 |          | BALL                                           | A01AC08   | 8                                                                                                                                                                | 8    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO                                | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 90                                                                                 | U04 |          | BALL                                           | B01AC01   | 0.5                                                                                                                                                              | 2    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                 | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 100                                                                                | U04 | VA5      | BALL                                           | B01AC01   | 3                                                                                                                                                                | 3    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                 | 0                           | 0                                                                                                                                  | FA3   | FLAT FACE FLANGES & FULL FACE GASKETS SHALL ONLY BE USED WHEN MATING WITH FLAT FACE EQUIPMENT.                                          |    |            |            |                   |                |        |
| 110                                                                                | U04 |          | BALL                                           | B01AC02   | 0.5                                                                                                                                                              | 2    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO                                | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 120                                                                                | U04 | VA5      | BALL                                           | B01AC02   | 3                                                                                                                                                                | 3    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO                                | 0                           | 0                                                                                                                                  | FA8   | USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.                                                                                   |    |            |            |                   |                |        |
| 130                                                                                | U04 | VA5      | BALL                                           | B01AC03   | 4                                                                                                                                                                | 4    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                 | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 140                                                                                | U04 | VA5      | BALL                                           | B01AC04   | 4                                                                                                                                                                | 4    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO                                | 0                           | 0                                                                                                                                  | FF5   | FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES                                                                          |    |            |            |                   |                |        |
| 150                                                                                | U04 | VA5      | BALL                                           | B01AC07   | 6                                                                                                                                                                | 8    | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO                                 | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 160                                                                                | U04 | VA5      | BALL                                           | B01AC07   | 10                                                                                                                                                               | 24   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO                                 | 0                           | 0                                                                                                                                  | FS1   | USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.                                                         |    |            |            |                   |                |        |
| 170                                                                                | U04 | VA5      | BALL                                           | B01AC08   | 6                                                                                                                                                                | 24   | BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO                                | 0                           | 0                                                                                                                                  | GA6   | SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117                                                                                 |    |            |            |                   |                |        |
| 180                                                                                | U04 | FS1      | BALL                                           | B03AC01   | 0.5                                                                                                                                                              | 3    | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO                                 | 0                           | 0                                                                                                                                  | ND9   | USE ONLY WITH CONSENT OF MATERIALS ENGINEER.                                                                                            |    |            |            |                   |                |        |
| 190                                                                                | U04 | FS1      | BALL                                           | B03AC02   | 0.5                                                                                                                                                              | 2    | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO                                | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 200                                                                                | U04 | FS1      | BALL                                           | A03AC01   | 3                                                                                                                                                                | 3    | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO                                 | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 210                                                                                | U04 | FS1      | BALL                                           | A03AC02   | 3                                                                                                                                                                | 3    | BALL 300# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO                                | 0                           | 0                                                                                                                                  | SB6   | USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.                                                     |    |            |            |                   |                |        |
| 220                                                                                |     |          |                                                |           |                                                                                                                                                                  |      | ***** BUTTERFLY VALVE *****                                                                    |                             |                                                                                                                                    | UD7   | USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST                                  |    |            |            |                   |                |        |
| 230                                                                                |     |          | BUTTERFLY                                      | T01AC61   | 10                                                                                                                                                               | 24   | BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO                      | 0                           | 0                                                                                                                                  | UG3   | PTFE GASKET ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING                                                   |    |            |            |                   |                |        |
| 240                                                                                |     |          | BUTTERFLY                                      | T01JC61   | 30                                                                                                                                                               | 30   | BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO ASME B16.47 SERIES A | 0                           | 0                                                                                                                                  | UG4   | BOLTS ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING                                                         |    |            |            |                   |                |        |
| 250                                                                                |     |          |                                                |           |                                                                                                                                                                  |      | ***** GATE VALVES *****                                                                        |                             |                                                                                                                                    | UJ2   | USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.                       |    |            |            |                   |                |        |
| 260                                                                                |     |          |                                                |           |                                                                                                                                                                  |      |                                                                                                |                             |                                                                                                                                    | UK9   | DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.                                                                                           |    |            |            |                   |                |        |
| 260                                                                                | U04 |          | GATE                                           | G01AC48   | 0.5                                                                                                                                                              | 1.5  | GATE 150# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW                                               | 0                           | 0                                                                                                                                  | VA5   | USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS                                                                 |    |            |            |                   |                |        |
| 270                                                                                |     |          | GATE                                           | G01AC41   | 2                                                                                                                                                                | 8    | GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW                                                    | 0                           | 0                                                                                                                                  | VAR   | ONLY TO BE USED WHEN SPECIFIED ON THE P&ID.                                                                                             |    |            |            |                   |                |        |
| 280                                                                                |     | VAR      | GATE                                           | G01AC41   | 10                                                                                                                                                               | 10   | GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW                                                    | 0                           | 0                                                                                                                                  | VB2   | SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.                                                                    |    |            |            |                   |                |        |
| 290                                                                                | U04 | VAR      | GATE                                           | G01AC41   | 14                                                                                                                                                               | 14   | GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW                                                    | 0                           | 0                                                                                                                                  | VC1   | SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL. |    |            |            |                   |                |        |
| 300                                                                                | U04 |          | GATE                                           | G03AC48   | 0.75                                                                                                                                                             | 1.5  | GATE 300# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW                                               | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 310                                                                                | U04 |          | GATE                                           | G03AC41   | 2                                                                                                                                                                | 3    | GATE 300# RF LTCS A352 LCC TRIM 16 WEDGE HW                                                    | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 320                                                                                |     |          |                                                |           |                                                                                                                                                                  |      | ***** GLOBE VALVES *****                                                                       |                             |                                                                                                                                    |       |                                                                                                                                         |    |            |            |                   |                |        |
| 330                                                                                |     |          |                                                |           |                                                                                                                                                                  |      |                                                                                                |                             |                                                                                                                                    |       |                                                                                                                                         |    |            |            |                   |                |        |
| 330                                                                                | U04 |          | GLOBE                                          | L01AC48   | 0.75                                                                                                                                                             | 1.5  | GLOBE 150# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW                                               | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 340                                                                                |     |          | GLOBE                                          | L01AC41   | 2                                                                                                                                                                | 6    | GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y HW                                                    | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 350                                                                                |     |          | GLOBE                                          | L01AC42   | 8                                                                                                                                                                | 10   | GLOBE 150# RF LTCS A352 LCC TRIM 16 OS&Y GO                                                    | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 360                                                                                |     |          |                                                |           |                                                                                                                                                                  |      | ***** CHECK VALVES *****                                                                       |                             |                                                                                                                                    |       |                                                                                                                                         |    |            |            |                   |                |        |
| 370                                                                                |     |          |                                                |           |                                                                                                                                                                  |      |                                                                                                |                             |                                                                                                                                    |       |                                                                                                                                         |    |            |            |                   |                |        |
| 370                                                                                | U04 | VC1      | CHECK                                          | C01AC50   | 0.5                                                                                                                                                              | 1.5  | CHECK PISTON 150# RF LTCS A350 LF2 CL.1 TRIM 16                                                | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 380                                                                                | U04 |          | CHECK                                          | C01AC45   | 2                                                                                                                                                                | 6    | CHECK DUAL WAFER LUG 150# RF LTCS A352 LCC TRIM 16 THROU BOLTED                                | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 390                                                                                | U04 |          | CHECK                                          | C01AC52   | 8                                                                                                                                                                | 24   | CHECK DUAL DBL FLG 150# RF LTCS A352 LCC TRIM 16                                               | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 400                                                                                |     |          |                                                |           |                                                                                                                                                                  |      | ***** PIPE *****                                                                               |                             |                                                                                                                                    |       |                                                                                                                                         |    |            |            |                   |                |        |
| 410                                                                                |     |          |                                                |           |                                                                                                                                                                  |      |                                                                                                |                             |                                                                                                                                    |       |                                                                                                                                         |    |            |            |                   |                |        |
| 410                                                                                |     |          | PIPE                                           | 5358508   | 0.75                                                                                                                                                             | 1.5  | PIPE SCH XS SMLS LTCS A333 GR.6.PE                                                             | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 420                                                                                |     |          | PIPE                                           | 5358514   | 2                                                                                                                                                                | 2.5  | PIPE SCH XS SMLS LTCS A333 GR.6 BE                                                             | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 430                                                                                |     |          | PIPE                                           | 5358505   | 3                                                                                                                                                                | 14   | PIPE SCH STD SMLS LTCS A333 GR.6 BE                                                            | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 440                                                                                |     |          | PIPE                                           | 5358501   | 16                                                                                                                                                               | 16   | PIPE SCH 20 SMLS LTCS A333 GR.6 BE                                                             | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 450                                                                                |     |          | PIPE                                           | 5358597   | 18                                                                                                                                                               | 30   | PIPE SCH 20 EFW LTCS A671 CC60 CL.22 BE                                                        | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 460                                                                                |     |          | PIPE                                           | 5358577   | 40                                                                                                                                                               | 40   | PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE                                                        | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 470                                                                                |     | 810      | PIPE                                           | 5358512   | 0.5                                                                                                                                                              | 0.5  | PIPE SCH 160 SMLS LTCS A333 GR.6 PE                                                            | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 480                                                                                | U04 | UK9      | PIPE                                           | 5358511   | 0.5                                                                                                                                                              | 0.75 | PIPE SCH XXS SMLS LTCS A333 GR.6 PE                                                            | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 490                                                                                |     |          |                                                |           |                                                                                                                                                                  |      | ***** FITTINGS*****                                                                            |                             |                                                                                                                                    |       |                                                                                                                                         |    |            |            |                   |                |        |
| 500                                                                                |     |          |                                                |           |                                                                                                                                                                  |      |                                                                                                |                             |                                                                                                                                    |       |                                                                                                                                         |    |            |            |                   |                |        |
| 500                                                                                | U04 |          | ELBOW                                          | 5392600   | 0.5                                                                                                                                                              | 2.5  | ELL BW 90 DEG LR SCH XS LTCS A420 WPL6                                                         | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 510                                                                                |     |          | ELBOW                                          | 5392438   | 3                                                                                                                                                                | 14   | ELL BW 90 DEG LR SCH STD LTCS A420 WPL6                                                        | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 520                                                                                | U04 | UK9      | ELBOW                                          | 5392815   | 0.5                                                                                                                                                              | 0.75 | ELL BW 90 DEG LR SCH XXS LTCS A420 WPL6                                                        | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 530                                                                                |     |          | ELBOW                                          | 5392276   | 16                                                                                                                                                               | 16   | ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6                                                         | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 540                                                                                |     |          | ELBOW                                          | 5392280   | 18                                                                                                                                                               | 30   | ELL BW 90 DEG LR SCH 20 LTCS A420 WPL6-W                                                       | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 550                                                                                |     |          | ELBOW                                          | 5392497   | 40                                                                                                                                                               | 40   | ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W                                                       | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 560                                                                                |     |          | ELBOW                                          | 5392606   | 0.75                                                                                                                                                             | 2    | ELL BW 45 DEG LR SCH XS LTCS A420 WPL6                                                         | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 570                                                                                | U04 | UK9      | ELBOW                                          | 5392831   | 0.5                                                                                                                                                              | 0.75 | ELL BW 45 DEG LR SCH XXS LTCS A420 WPL6                                                        | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 580                                                                                |     |          | ELBOW                                          | 5392444   | 3                                                                                                                                                                | 14   | ELL BW 45 DEG LR SCH STD LTCS A420 WPL6                                                        | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
| 590                                                                                |     |          | ELBOW                                          | 5392282   | 16                                                                                                                                                               | 16   | ELL BW 45 DEG SCH 20 LTCS A420 WPL6                                                            | 0                           | 0                                                                                                                                  |       |                                                                                                                                         |    |            |            |                   |                |        |
|                                                                                    |     |          |                                                |           |                                                                                                                                                                  |      |                                                                                                |                             |                                                                                                                                    |       | DATE                                                                                                                                    | BY | REV        | SHEET      | PIPING LINE CLASS |                |        |
|  |     |          | TENGIZCHEVROIL<br>TCO<br>TENGIZ,<br>KAZAKHSTAN |           | CONSTRUCTION<br>ALL SIZES - BUTT WELD<br>PWHT FOR WT > 19 MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE |      | CORR.ALLOW.<br>CORR. ALLOW. 3 MM                                                               |                             | SERVICE<br>SOUR SERVICE - NACE MR0175/ ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO |       | MATERIAL<br>MATERIAL: LTCS<br>FLANGE RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 343 °C                                                  |    | 12/16/2024 | BY<br>OZAC | U04               | Page 1<br>of 4 | 150H25 |

| SEQ NO | REV | NOTE NO. | ITEM NAME  | ITEM CODE | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                                                               | REFERENCE / LARGE END SIZES |     | NOTES |
|--------|-----|----------|------------|-----------|------------------------------|------|-------------------------------------------------------------------------------------------|-----------------------------|-----|-------|
| 600    |     |          | ELBOW      | 5392281   | 18                           | 30   | ELL BW 45 DEG LR SCH 20 LTCS A420 WPL6-W                                                  | 0                           | 0   |       |
| 610    |     |          | ELBOW      | 5392608   | 40                           | 40   | ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W                                                  | 0                           | 0   |       |
| 620    |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 620    |     |          | INDUC BEND | 5800095   | 30                           | 30   | INDUCTION BEND 30 DEG. X.5D RADIUS SCH 20 ASTM A671 CC60 CL.22 - C/W 1.5D TANGENT ENDS BE | 0                           | 0   |       |
| 630    |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 630    | U04 |          | TEE        | 5392609   | 0.5                          | 2    | TEE BW SCH XS LTCS A420 WPL6                                                              | 0                           | 0   |       |
| 640    | U04 | UK9      | TEE        | 5392879   | 0.5                          | 0.75 | TEE SCH XXS LTCS A420 WPL6                                                                | 0                           | 0   |       |
| 650    |     |          | TEE        | 5392447   | 3                            | 14   | TEE BW SCH STD LTCS A420 WPL6                                                             | 0                           | 0   |       |
| 660    |     |          | TEE        | 5392285   | 16                           | 16   | TEE BW SCH 20 LTCS A420 WPL6                                                              | 0                           | 0   |       |
| 670    |     |          | TEE        | 5392295   | 18                           | 30   | TEE BW SCH 20 LTCS A420 WPL6-W                                                            | 0                           | 0   |       |
| 680    | U04 |          | TEE RED    | 5392661   | 0.5                          | 1.5  | TEE RED BW SCH XS LTCS A420 WPL6                                                          | 0.75                        | 2   |       |
| 690    |     |          | TEE RED    | 5392454   | 1.5                          | 2    | TEE RED BW SCH STD/XS LTCS A420 WPL6                                                      | 3                           | 4   |       |
| 700    |     |          | TEE RED    | 5392445   | 3                            | 12   | TEE RED BW SCH STD LTCS A420 WPL6                                                         | 4                           | 14  |       |
| 710    |     |          | TEE RED    | 5392289   | 6                            | 14   | TEE RED BW SCH 20/STD LTCS A420 WPL6                                                      | 16                          | 16  |       |
| 720    |     |          | TEE RED    | 5392296   | 8                            | 14   | TEE RED BW SCH 20/STD LTCS A420 WPL6-W                                                    | 18                          | 30  |       |
| 730    |     |          | TEE RED    | 5392292   | 16                           | 24   | TEE RED BW SCH 20 LTCS A420 WPL6-W                                                        | 18                          | 30  |       |
| 740    |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 740    |     |          | CROSS      | 5416601   | 4                            | 4    | CROSS SCH STD LTCS ASTM A420 WPL6                                                         | 0                           | 0   |       |
| 750    |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 750    |     |          | CAP        | 5392610   | 0.75                         | 2    | CAP BW SCH XS LTCS A420 WPL6                                                              | 0                           | 0   |       |
| 760    |     |          | CAP        | 5392448   | 3                            | 14   | CAP BW SCH STD LTCS A420 WPL6                                                             | 0                           | 0   |       |
| 770    |     |          | CAP        | 5392286   | 16                           | 30   | CAP BW SCH 20 LTCS A420 WPL6                                                              | 0                           | 0   |       |
| 780    |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 780    | U04 |          | RED CONC   | 5392458   | 0.5                          | 1.5  | REDUCER CONC SCH XS LTCS A420 WPL6                                                        | 0.75                        | 2   |       |
| 790    |     |          | RED CONC   | 5392452   | 1.5                          | 2.5  | REDUCER CONC SCH STD/XS LTCS A420 WPL6                                                    | 3                           | 4   |       |
| 800    |     |          | RED CONC   | 5392449   | 3                            | 12   | REDUCER CONC SCH STD LTCS A420 WPL6                                                       | 4                           | 14  |       |
| 810    |     |          | RED CONC   | 5392290   | 8                            | 14   | REDUCER CONC SCH 20/STD LTCS A420 WPL6                                                    | 16                          | 16  |       |
| 820    |     |          | RED CONC   | 5392297   | 10                           | 14   | REDUCER CONC SCH 20/STD LTCS A420 WPL6-W                                                  | 18                          | 20  |       |
| 830    |     |          | RED CONC   | 5392287   | 16                           | 24   | REDUCER CONC SCH 20 LTCS A420 WPL6-W                                                      | 18                          | 30  |       |
| 840    |     |          | RED CONC   | 5392512   | 30                           | 30   | REDUCER CONC SCH XS/20 LTCS A420 WPL6-W                                                   | 40                          | 40  |       |
| 850    | U04 |          | RED ECC    | 5392461   | 0.5                          | 1.5  | REDUCER ECC SCH XS LTCS A420 WPL6                                                         | 1                           | 2   |       |
| 860    | U04 |          | RED ECC    | 5392453   | 1.5                          | 2.5  | REDUCER ECC SCH STD/XS LTCS A420 WPL6                                                     | 3                           | 6   |       |
| 870    |     |          | RED ECC    | 5392450   | 3                            | 12   | REDUCER ECC SCH STD LTCS A420 WPL6                                                        | 4                           | 14  |       |
| 880    |     |          | RED ECC    | 5392291   | 8                            | 14   | REDUCER ECC SCH 20/STD LTCS A420 WPL6                                                     | 16                          | 16  |       |
| 890    |     |          | RED ECC    | 5392298   | 10                           | 14   | REDUCER ECC SCH 20/STD LTCS A420 WPL6-W                                                   | 18                          | 20  |       |
| 900    |     |          | RED ECC    | 5392288   | 16                           | 24   | REDUCER ECC SCH 20 LTCS A420 WPL6-W                                                       | 18                          | 30  |       |
| 910    |     |          | RED ECC    | 5392513   | 30                           | 30   | REDUCER ECC SCH XS/20 LTCS A420 WPL6-W                                                    | 40                          | 40  |       |
| 920    |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 920    | U04 |          | OLETS      | 5504001   | 0.75                         | 1.5  | NIPOFLANGE 150# RF SCH XS LTCS A350 LF2 CL.1                                              | 2                           | 2   |       |
| 930    | U04 |          | OLETS      | 5504002   | 0.75                         | 2    | NIPOFLANGE 150# RF SCH STD/XS LTCS A350 LF2 CL.1                                          | 3                           | 14  |       |
| 940    | U04 |          | OLETS      | 5504016   | 0.75                         | 2    | NIPOFLANGE 150# RF SCH 20/XS LTCS A350 LF2 CL.1                                           | 16                          | 24  |       |
| 950    | U04 |          | OLETS      | 5504020   | 0.75                         | 1.5  | NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1                                              | 1.5                         | 2   |       |
| 960    | U04 |          | OLETS      | 5504021   | 0.75                         | 1.5  | NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1                                          | 3                           | 14  |       |
| 970    | U04 | UJ2      | OLETS      | 5504025   | 1.5                          | 1.5  | NIPOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1                                         | 4                           | 14  |       |
| 980    | U04 | UJ2      | OLETS      | 5504019   | 1.5                          | 1.5  | NIPOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1                                          | 16                          | 24  |       |
| 990    | U04 | FF5      | OLETS      | 5504500   | 0.75                         | 1.5  | NIPOFLANGE FLAT BOTTOM 150# RF SCH XS LTCS A350 LF2 CL.1                                  | 0                           | 0   |       |
| 1000   |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 1000   | U04 |          | OLETS      | 5505335   | 0.75                         | 1    | WELDOLET SCH XS LTCS A350 LF2 CL.1                                                        | 2                           | 2   |       |
| 1010   | U04 |          | OLETS      | 5505179   | 0.75                         | 2.5  | WELDOLET SCH STD/XS LTCS A350 LF2 CL.1                                                    | 3                           | 14  |       |
| 1020   | U04 |          | OLETS      | 5505572   | 0.75                         | 2    | WELDOLET SCH 20/XS LTCS A350 LF2 CL.1                                                     | 16                          | 30  |       |
| 1030   | U04 |          | OLETS      | 5505164   | 3                            | 4    | WELDOLET SCH STD LTCS A350 LF2 CL.1                                                       | 8                           | 14  |       |
| 1040   | U04 |          | OLETS      | 5505164   | 3                            | 3    | WELDOLET SCH STD LTCS A350 LF2 CL.1                                                       | 6                           | 6   |       |
| 1050   | U04 |          | OLETS      | 5505571   | 3                            | 10   | WELDOLET SCH 20/STD LTCS A350 LF2 CL.1                                                    | 16                          | 30  |       |
| 1060   | U04 |          | OLETS      | 5504374   | 0.75                         | 1    | WELDOLET FLAT BOTTOM SCH XS LTCS A350 LF2 CL.1                                            | 0                           | 0   |       |
| 1070   | U04 |          | OLETS      | 5504374   | 2                            | 2    | WELDOLET FLAT BOTTOM SCH XS LTCS A350 LF2 CL.1                                            | 0                           | 0   |       |
| 1080   |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 1080   | U04 | UJ2      | ELBOFLG    | 5516625   | 1.5                          | 1.5  | ELBOFLANGE 300# RF SCH STD/160 LTCS A350 LF2 CL.1                                         | 4                           | 14  |       |
| 1090   | U04 | UJ2      | ELBOFLG    | 5516619   | 1.5                          | 1.5  | ELBOFLANGE 300# RF SCH 20/160 LTCS A350 LF2 CL.1                                          | 16                          | 24  |       |
| 1100   |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 1100   | U04 | ND9      | RED CONC   | 5544094   | 0.75                         | 0.75 | SWAGE CONC SCH XS/160 LTCS A420 WPL6 PLE/TSE                                              | 1                           | 1   |       |
| 1110   |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 1110   |     |          |            |           |                              |      | ***** OLETS *****                                                                         |                             |     |       |
| 1120   |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 1120   | U04 |          | OLETS      | 5533072   | 0.75                         | 1.5  | SOCKOLET 3000# LTCS A350 LF2 CL.1                                                         | 2                           | 24  |       |
| 1130   | U04 |          | OLETS      | 5533232   | 0.75                         | 1.5  | SOCKOLET FLAT BOTTOM 3000# LTCS A350 LF2 CL.1                                             | 0                           | 0   |       |
| 1140   |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 1140   | U04 | UD7      | OLETS      | 5514772   | 0.75                         | 0.75 | THREDOLET 3000# LTCS A350 LF2 CL.1                                                        | 2                           | 24  |       |
| 1150   |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 1150   |     |          |            |           |                              |      | ***** COUPLINGS, UNIONS & PLUGS *****                                                     |                             |     |       |
| 1160   |     |          |            |           |                              |      |                                                                                           |                             |     |       |
| 1160   |     |          | COUPLING   | 5478705   | 0.75                         | 1.5  | COUPLING 3000# SW LTCS A 350 LF2                                                          | 0                           | 0   |       |
| 1170   | U04 |          | TEE RED    | 5478709   | 0.75                         | 1    | COUPLING RED 3000# SW LTCS A350 LF2 CL.1                                                  | 1                           | 1.5 |       |
| 1180   |     |          |            |           |                              |      |                                                                                           |                             |     |       |

|  | <b>TENGIZCHEVROI<br/>TCO<br/>TENGIZ,<br/>KAZAKHSTAN</b> | CONSTRUCTION                                                                                                                                     | CORR.ALLOW.       | SERVICE                                                                                                                 | MATERIAL                                                                   | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|                                                                                    |                                                         | ALL SIZES - BUTT WELD<br>PWHT FOR WT > 19 MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW. 3 MM | SOUR SERVICE - NACE MR0175/ ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>FLANGE RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 343 °C | 12/16/2024 | BY<br>OZAC | U04 | Page 2<br>of 4 | <b>150H25</b>     |

| SEQ NO                                                                             | REV | NOTE NO. | ITEM NAME                                                                                                                                        | ITEM CODE | SIZE RANGE / SMALL END SIZES |      | DESCRIPTION                                                                                                             | REFERENCE / LARGE END SIZES                                                |     | NOTES      |            |     |                |                   |
|------------------------------------------------------------------------------------|-----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------------------------|------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----|------------|------------|-----|----------------|-------------------|
| 1180                                                                               | U04 |          | UNION                                                                                                                                            | 5478706   | 0.75                         | 1.5  | UNION 3000# SW LTCS A350 LF2 CL.1                                                                                       | 0                                                                          | 0   |            |            |     |                |                   |
| 1190                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1190                                                                               | U04 | UD7      | FITTING SCRD                                                                                                                                     | 5491303   | 0.75                         | 0.75 | PLUG ROUND HEAD SCRD LTCS A350 LF2 CL.1                                                                                 | 0                                                                          | 0   |            |            |     |                |                   |
| 1200                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1200                                                                               |     | FS1      |                                                                                                                                                  |           |                              |      | ***** FLANGES *****                                                                                                     |                                                                            |     |            |            |     |                |                   |
| 1210                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1210                                                                               | U04 | FA8      | TEE RED                                                                                                                                          | 5599692   | 0.5                          | 0.5  | FLG SCRD (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1                                                                     | 0.75                                                                       | 1.5 |            |            |     |                |                   |
| 1220                                                                               | U04 | FA8      | TEE RED                                                                                                                                          | 5599703   | 0.5                          | 0.5  | FLG SCRD (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1                                                                     | 0.75                                                                       | 2   |            |            |     |                |                   |
| 1230                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1230                                                                               |     |          | FLANGE                                                                                                                                           | 5851200   | 0.5                          | 2    | FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE                                                                              | 0                                                                          | 0   |            |            |     |                |                   |
| 1240                                                                               |     |          | FLANGE                                                                                                                                           | 5851203   | 0.5                          | 2    | FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE                                                                              | 0                                                                          | 0   |            |            |     |                |                   |
| 1250                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1250                                                                               | U04 |          | BLIND                                                                                                                                            | 5615566   | 0.5                          | 24   | FLG BLIND 150# RF LTCS A350 LF2 CL.1                                                                                    | 0                                                                          | 0   |            |            |     |                |                   |
| 1260                                                                               | U04 |          | BLIND                                                                                                                                            | 5615570   | 30                           | 30   | FLG BLIND 150# RF LTCS A350 LF2 CL.1 B16.47 SERIES A                                                                    | 0                                                                          | 0   |            |            |     |                |                   |
| 1270                                                                               | U04 |          | BLIND                                                                                                                                            | 5615576   | 0.5                          | 24   | FLG BLIND 300# RF LTCS A350 LF2 CL.1                                                                                    | 0                                                                          | 0   |            |            |     |                |                   |
| 1280                                                                               | U04 |          | BLIND                                                                                                                                            | 5615598   | 0.5                          | 24   | FLG BLIND 600# RF LTCS A350 LF2 CL.1                                                                                    | 0                                                                          | 0   |            |            |     |                |                   |
| 1290                                                                               | U04 |          | BLIND                                                                                                                                            | 5615616   | 2                            | 2    | FLG BLIND 1500# RF LTCS A350 LF2 CL.1                                                                                   | 0                                                                          | 0   |            |            |     |                |                   |
| 1300                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556264   | 0.5                          | 2.5  | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS                                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1310                                                                               | U04 |          | FLANGE                                                                                                                                           | 5557393   | 0.75                         | 2    | FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS                                                                               | 0                                                                          | 0   |            |            |     |                |                   |
| 1320                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556476   | 0.5                          | 2    | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS                                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1330                                                                               | U04 |          | FLANGE                                                                                                                                           | 5557036   | 0.75                         | 2    | FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS                                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1340                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556217   | 2                            | 2    | FLG WN 150# FF LTCS A350 LF2 CL.1 SCH XS                                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1350                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556261   | 3                            | 14   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD                                                                               | 0                                                                          | 0   |            |            |     |                |                   |
| 1360                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556216   | 3                            | 3    | FLG WN 150# FF LTCS A350 LF2 CL.1 SCH STD                                                                               | 0                                                                          | 0   |            |            |     |                |                   |
| 1370                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556473   | 3                            | 14   | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH STD                                                                               | 0                                                                          | 0   |            |            |     |                |                   |
| 1380                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556456   | 8                            | 10   | FLG WN 300# FF LTCS A350 LF2 CL.1 SCH STD                                                                               | 0                                                                          | 0   |            |            |     |                |                   |
| 1390                                                                               | U04 |          | FLANGE                                                                                                                                           | 5557033   | 3                            | 14   | FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD                                                                               | 0                                                                          | 0   |            |            |     |                |                   |
| 1400                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556265   | 16                           | 24   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 20                                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1410                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556474   | 16                           | 24   | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 20                                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1420                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556468   | 30                           | 30   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH 20 B16.47 SERIES A                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1430                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556469   | 30                           | 30   | FLG WN 300# RF LTCS A350 LF2 CL.1 SCH 20 B16.47 SERIES A                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1440                                                                               | U04 |          | FLANGE                                                                                                                                           | 5556266   | 40                           | 40   | FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS B16.47 SERIES A                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1450                                                                               | U04 |          | FLANGE                                                                                                                                           | 5555935   | 3                            | 3    | FLG WN 900# RF LTCS A350 LF2 CL.1 SCH STD                                                                               | 0                                                                          | 0   |            |            |     |                |                   |
| 1460                                                                               | U04 |          | FLANGE                                                                                                                                           | 5555936   | 2                            | 2    | FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XS                                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1470                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1470                                                                               |     |          |                                                                                                                                                  |           |                              |      | ***** ORIFICE FLANGES *****                                                                                             |                                                                            |     |            |            |     |                |                   |
| 1480                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1480                                                                               | U04 |          | ORIFICE FLG                                                                                                                                      | 5623947   | 2                            | 2    | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS                                                                           | 0                                                                          | 0   |            |            |     |                |                   |
| 1490                                                                               | U04 |          | ORIFICE FLG                                                                                                                                      | 5623944   | 3                            | 14   | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH STD                                                                          | 0                                                                          | 0   |            |            |     |                |                   |
| 1500                                                                               | U04 |          | ORIFICE FLG                                                                                                                                      | 5623941   | 16                           | 24   | WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH 20                                                                           | 0                                                                          | 0   |            |            |     |                |                   |
| 1510                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1510                                                                               |     |          |                                                                                                                                                  |           |                              |      | ***** PIPE NIPPLES *****                                                                                                |                                                                            |     |            |            |     |                |                   |
| 1520                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1520                                                                               | U04 | 810      |                                                                                                                                                  |           |                              |      | NIPPLE PBE CUT FROM PIPE                                                                                                |                                                                            |     |            |            |     |                |                   |
| 1530                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1530                                                                               |     | FS1      |                                                                                                                                                  |           |                              |      | ***** GASKETS *****                                                                                                     |                                                                            |     |            |            |     |                |                   |
| 1540                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1540                                                                               |     |          | GASKET                                                                                                                                           | 5673043   | 0.5                          | 24   | GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR                                                                              | 0                                                                          | 0   |            |            |     |                |                   |
| 1550                                                                               |     |          | GASKET                                                                                                                                           | 5674011   | 30                           | 40   | GASKET 150# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A                                                             | 0                                                                          | 0   |            |            |     |                |                   |
| 1560                                                                               |     |          | GASKET                                                                                                                                           | 5664562   | 2                            | 3    | GASKET 150# FF RING EPDM 3MM THK                                                                                        | 0                                                                          | 0   |            |            |     |                |                   |
| 1570                                                                               |     | FA3      | GASKET                                                                                                                                           | 5673033   | 0.5                          | 24   | GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR                                                                              | 0                                                                          | 0   |            |            |     |                |                   |
| 1580                                                                               |     |          | GASKET                                                                                                                                           | 5674032   | 30                           | 30   | GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A                                                             | 0                                                                          | 0   |            |            |     |                |                   |
| 1590                                                                               |     | FA3      | GASKET                                                                                                                                           | 5664563   | 8                            | 10   | GASKET 300# FF RING EPDM 3MM THICK                                                                                      | 0                                                                          | 0   |            |            |     |                |                   |
| 1600                                                                               | U04 |          | GASKET                                                                                                                                           | 5673103   | 0.75                         | 6    | GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR                                                                              | 0                                                                          | 0   |            |            |     |                |                   |
| 1610                                                                               | U04 |          | GASKET                                                                                                                                           | 5673103   | 16                           | 16   | GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR                                                                              | 0                                                                          | 0   |            |            |     |                |                   |
| 1620                                                                               | U04 | ND9      | GASKET                                                                                                                                           | 5673060   | 2                            | 2    | GASKET 150# MONEL 400 SPIRAL WOUND GRAPHITE FILLED                                                                      | 0                                                                          | 0   |            |            |     |                |                   |
| 1630                                                                               | U04 | UG3      | GASKET                                                                                                                                           | 5664201   | 10                           | 10   | GASKET 150# RF FLAT RING PTFE                                                                                           | 0                                                                          | 0   |            |            |     |                |                   |
| 1640                                                                               | U04 | UG3      | GASKET                                                                                                                                           | 5664201   | 16                           | 16   | GASKET 150# RF FLAT RING PTFE                                                                                           | 0                                                                          | 0   |            |            |     |                |                   |
| 1650                                                                               | U04 |          | GASKET                                                                                                                                           | 5672251   | 1                            | 2    | GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR                                                                             | 0                                                                          | 0   |            |            |     |                |                   |
| 1660                                                                               | U04 | GA6      | GASKET                                                                                                                                           | 5673017   | 2                            | 3    | GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.5                                                                       | 0                                                                          | 0   |            |            |     |                |                   |
| 1670                                                                               | U04 |          | GASKET                                                                                                                                           | 5673117   | 2                            | 3    | GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR                                                                              | 0                                                                          | 0   |            |            |     |                |                   |
| 1680                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1680                                                                               |     | B1       |                                                                                                                                                  |           |                              |      | ***** BOLTING *****                                                                                                     |                                                                            |     |            |            |     |                |                   |
| 1690                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1690                                                                               | U04 |          | BOLTING                                                                                                                                          | 5675770   | 0.5                          | 99   | STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS                                                                       | 0                                                                          | 0   |            |            |     |                |                   |
| 1700                                                                               | U04 | UG4      | BOLTING                                                                                                                                          | 5675768   | 0.5                          | 99   | STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS                                                          | 0                                                                          | 0   |            |            |     |                |                   |
| 1710                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1710                                                                               |     | SB6      | BOLTING                                                                                                                                          | 5675675   | 12                           | 24   | JACK SCREW HEX HEAD A320 GR L7M                                                                                         | 0                                                                          | 0   |            |            |     |                |                   |
| 1720                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1720                                                                               |     | FS1      |                                                                                                                                                  |           |                              |      | **** BLINDS & SPACERS ****                                                                                              |                                                                            |     |            |            |     |                |                   |
| 1730                                                                               |     |          |                                                                                                                                                  |           |                              |      |                                                                                                                         |                                                                            |     |            |            |     |                |                   |
| 1730                                                                               |     |          | SPEC BLIND                                                                                                                                       | 5749580   | 0.75                         | 14   | SPECTACLE BLIND 150# RF LTCS A 516 GR.70                                                                                | 0                                                                          | 0   |            |            |     |                |                   |
| 1740                                                                               |     |          | SPEC BLIND                                                                                                                                       | 5749581   | 1                            | 10   | SPECTACLE BLIND 300# RF LTCS A516 GR.70                                                                                 | 0                                                                          | 0   |            |            |     |                |                   |
|  |     |          | CONSTRUCTION                                                                                                                                     |           | CORR.ALLOW.                  |      | SERVICE                                                                                                                 | MATERIAL                                                                   |     | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|                                                                                    |     |          | ALL SIZES - BUTT WELD<br>PWHT FOR WT > 19 MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE |           | CORR. ALLOW. 3 MM            |      | SOUR SERVICE - NACE MR0175/ ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>FLANGE RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 343 °C |     | 12/16/2024 | BY<br>OZAC | U04 | Page 3<br>of 4 | 150H25            |

| SEQ NO | REV | NOTE NO. | ITEM NAME  | ITEM CODE | SIZE RANGE / SMALL END SIZES |    | DESCRIPTION                                           | REFERENCE / LARGE END SIZES |   | NOTES |
|--------|-----|----------|------------|-----------|------------------------------|----|-------------------------------------------------------|-----------------------------|---|-------|
| 1750   | U04 |          | SPEC BLIND | 5749587   | 0.75                         | 8  | SPECTACLE BLIND 600# RF LTCS A516-GR70                | 0                           | 0 |       |
| 1760   |     |          |            |           |                              |    |                                                       |                             |   |       |
| 1760   |     |          | PADDLE     | 5750023   | 16                           | 24 | PADDLE BLANK 150# RF LTCS A516 GR.70                  | 0                           | 0 |       |
| 1770   |     |          | PADDLE     | 5750100   | 12                           | 24 | PADDLE BLANK 300# RF LTCS A516 GR.70                  | 0                           | 0 |       |
| 1780   |     |          | PADDLE     | 5750028   | 30                           | 30 | PADDLE BLANK 300# RF LTCS A516 GR70 B16.47 SERIES A   | 0                           | 0 |       |
| 1790   | U04 |          | PADDLE     | 5750026   | 30                           | 30 | PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A   | 0                           | 0 |       |
| 1800   | U04 |          | PADDLE     | 5750026   | 40                           | 40 | PADDLE BLANK 150# RF LTCS A516 GR70 B16.47 SERIES A   | 0                           | 0 |       |
| 1810   | U04 |          | PADDLE     | 5750196   | 30                           | 30 | PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A | 0                           | 0 |       |
| 1820   | U04 |          | PADDLE     | 5750196   | 40                           | 40 | PADDLE SPACER 150# RF LTCS A516 GR.70 B16.47 SERIES A | 0                           | 0 |       |
| 1830   |     |          | PADDLE     | 5750240   | 16                           | 24 | PADDLE SPACER 150# RF LTCS A516 GR.70                 | 0                           | 0 |       |
| 1840   |     |          | PADDLE     | 5750200   | 12                           | 24 | PADDLE SPACER 300# RF LTCS A516 GR.70                 | 0                           | 0 |       |
| 1850   |     |          | PADDLE     | 5750198   | 30                           | 30 | PADDLE SPACER 300# RF LTCS A516 GR70 B16.47 SERIES A  | 0                           | 0 |       |
| 1860   |     |          |            |           |                              |    |                                                       |                             |   |       |
| 1870   |     |          |            |           |                              |    |                                                       |                             |   |       |
| 1870   | U04 | 823      | FLG GUARD  | 5558881   | 0.5                          | 10 | SAFETY SHIELD PTFE SUITABLE FOR 150# FLANGE           | 0                           | 0 |       |
| 1880   | U04 |          | FLG GUARD  | 5558882   | 0.75                         | 1  | SAFETY SHIELD PTFE SUITABLE FOR 300# FLANGE           | 0                           | 0 |       |
| 1890   | U04 |          | FLG GUARD  | 5558882   | 4                            | 4  | SAFETY SHIELD PTFE SUITABLE FOR 300# FLANGE           | 0                           | 0 |       |

|                                                                                    |                                                |                                                                                                                                                  |                   |                                                                                                                         |                                                                            |            |            |     |                |                   |
|------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------|------------|-----|----------------|-------------------|
|  | TENGIZCHEVROIL<br>TCO<br>TENGIZ,<br>KAZAKHSTAN | CONSTRUCTION                                                                                                                                     | CORR.ALLOW.       | SERVICE                                                                                                                 | MATERIAL                                                                   | DATE       | BY         | REV | SHEET          | PIPING LINE CLASS |
|                                                                                    |                                                | ALL SIZES - BUTT WELD<br>PWHT FOR WT > 19 MM<br>SEAL WELDING - FOR THREADED CONNECTIONS ONLY<br>PIPE BENDS - NONE<br>THREAD SEALANT - PTFE PASTE | CORR. ALLOW. 3 MM | SOUR SERVICE - NACE MR0175/ ISO 15156<br>DESIGN: ASME B16.5 MG 1.1<br>CODE: ASME B31.3<br>FLUID CODES - PIM-SU-5112-TCO | MATERIAL: LTCS<br>FLANGE RATING: 150 RF<br>TEMPERATURE RANGE: -40 / 343 °C | 12/16/2024 | BY<br>OZAC | U04 | Page 4<br>of 4 | 150H25            |